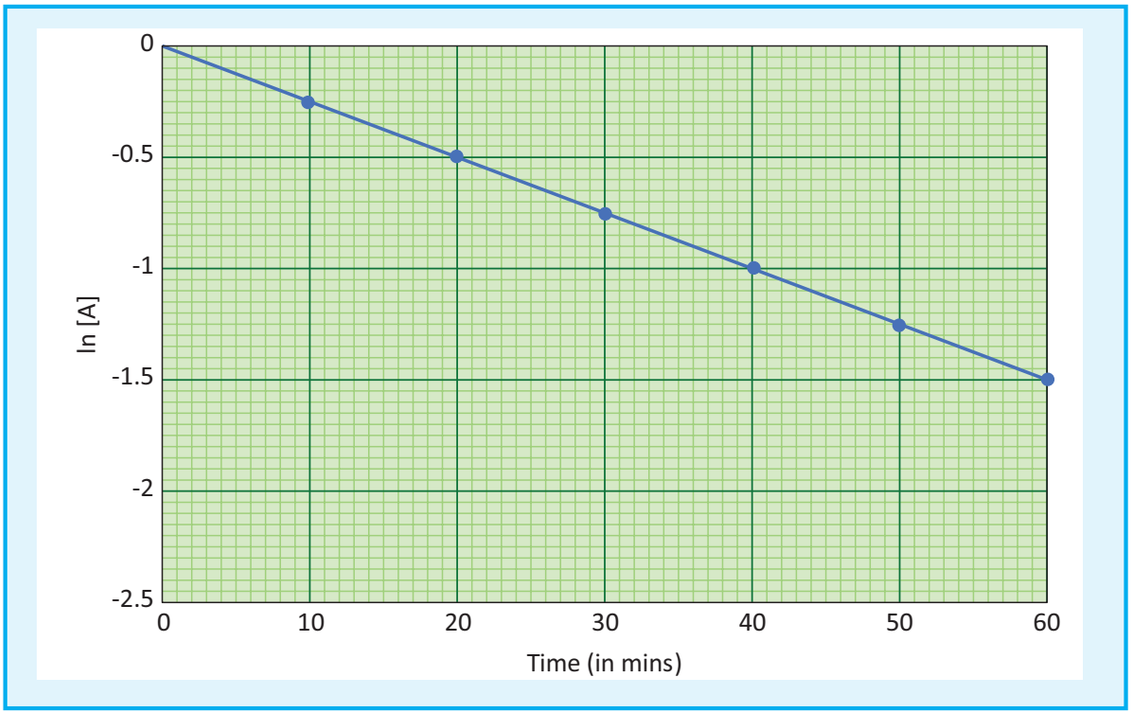


Fig: 7.3 A plot of ln[A] Vs t for a first order reaction, A ⟶ product with initial concentration of [A] = 1.00 M and k = 2.5 x$10^{-2}$$min^{-1}$.

$$2.303 \log\left(\frac{[A_0]}{[A]}\right) = kt$$

$$k = \frac{2.303}{t} \log\left(\frac{[A_0]}{[A]}\right) \text{ ----- (3)}$$

Equation (2) can be written in the form y = mx + c as below

$\ln[A_0] - \ln[A] = kt$

$\ln[A] = \ln[A_0] - kt$

$\Rightarrow y = c + mx$

If we follow the reaction by measuring the concentration of the reactants at regular time interval't', a plot of ln[A] against 't' yields a straight line with a negative slope.From this, the rate constant is calculated.

Examples for the first order reaction

(i) Decomposition of dinitrogen pentoxide

$$N_2O_5(g) \longrightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$$

(ii) Decomposition of thionylchloride; $SO_2Cl_2(l) \longrightarrow SO_2(g) + Cl_2(g)$

(iii) Decomposition of the H_2O_2 in aqueous solution; $H_2O_2(aq) \longrightarrow H_2O(l) + \frac{1}{2}O_2(g)$

(iv) Isomerisation of cyclopropane to propene.

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